

Logical Energy as Test-Time Compute: Scoring and Repairing Neural Outputs with a Differentiable Theorem-Proving Energy-Based Model

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Technical Report · Project ThermoLogic · Reproducible PyTorch prototype

Scope & honesty note. This is an engineering technical report / proof-of-concept, not a peer-reviewed claim of a new paradigm. It builds directly on established neuro-symbolic work (Logic Tensor Networks, DeepProbLog, Semantic Loss, Neural Theorem Provers, and analyses of differentiable fuzzy operators). Every number is produced by the accompanying open code with fixed seeds. The contribution is (i) a clean, fully-reproducible implementation, and (ii) two small original observations about *test-time energy repair* and how *reasoning depth maps to test-time compute*.

In plain terms. AI models often produce answers that break basic rules — an invalid configuration, a self-contradiction — with no built-in way to notice. ThermoLogic gives any answer a *score* for how much it breaks your rules (zero means it obeys all of them), then “rolls it downhill” to the nearest valid answer, using only the rules and no training data. This report shows the idea works on a small benchmark, that the score doubles as a label-free error detector, and that harder (deeper) reasoning takes proportionally more compute.

Abstract. Neural models trained by maximum likelihood have no internal mechanism forbidding logically inconsistent outputs. We present **Project ThermoLogic**, a compact PyTorch system that fuses Differentiable Theorem Proving (relaxing discrete rules into fuzzy-logic t-norms) with an Energy-Based Model: a belief state's energy is ~ 0 when it satisfies every rule and grows exponentially as rules are violated. A small MLP proposes probabilistic truth values; the energy acts as a differentiable prover. On an 11-atom benchmark with a genuine world-level train/test split we report six findings. (1) A plain supervised network matches ThermoLogic on raw accuracy (0.955 vs 0.970) but its outputs are logically inconsistent (mean energy 0.95); the energy layer's value is a **label-free consistency guarantee** (energy 0.001), not accuracy. (2) **Test-time energy repair** — gradient descent on the energy — turns an inconsistent output on unseen inputs into a valid one (0.935 $\rightarrow$ 0.971 accuracy; energy 0.90 $\rightarrow$ 0.0006). (3) In a proof chain, truth propagates **one hop at a time**: inference cost grows linearly with reasoning depth, and the t-norm sets the propagation speed (~ 3.4 descent steps/hop for Łukasiewicz vs ~ 1.0 for Gödel/Product). (4) Asked “why not just project onto the nearest valid state with a solver?”, we answer honestly: an exact projection **ties** on accuracy (0.969 vs 0.970) — the energy's edge is **linear scaling** (measured: enumerate-and-project cost grows ~ 20 orders of magnitude from a 3×3 Latin square to 9×9 Sudoku while repair time grows $\sim 30\times$) and **differentiability** (it can be trained through; a projection gives zero gradient). (5) That differentiability is a null for generalization with full labels, but as a semi-supervised loss on a *hard* target (parity) it lifts accuracy by up to **+26 points** when labels are scarce — and a head-to-head confirms this is a *distinct* mechanism from Semantic Loss (Xu et al., 2018), matching or beating it here and far more robust to the loss weight. (6) On an *external, non-circular* benchmark we did not author — order-4 Latin squares — the method **loses on both axes**: an exact solver dominates soft energy repair at runtime (100% vs 3.6% valid) and the logic loss gives no training-time lift, because those constraints are permutation-symmetric and carry no signal about the target. The transferable rule: reach for the differentiable logic when you need a *gradient*, *scale* past enumeration, or *scarce-label target signal* — and skip it when a solver fits. The mechanism ships as a small library that scores and repairs any model's structured output against hard rules at a controllable compute budget.

Keywords: neuro-symbolic AI, energy-based models, differentiable logic, t-norms, test-time compute, constraint satisfaction, guardrails, PyTorch.

1. Introduction

A maximum-likelihood objective rewards fitting the data distribution; it never assigns infinite cost to a self-contradictory output. Symbolic systems are the opposite — rigorously consistent but non-differentiable. Project ThermoLogic asks a narrow, testable question: *can a neural network's continuous, probabilistic beliefs be constrained — and repaired — by discrete logical rules, using only backpropagation?* The organizing metaphor is thermodynamic: logical validity corresponds to low energy (a ground state), a contradiction to high energy. Because the energy is differentiable, it can be minimized at training time (as a loss) and at test time (as an inference procedure that repairs a given output). (*The thermodynamic framing is an analogy for intuition, not literal physics: there is no entropy, and the “inverse temperature” β is simply a hyperparameter that sharpens the penalty.*)

Contributions. (1) A clean, reproducible DTP + EBM implementation with three t-norm families and residuated implications. (2) A label-free consistency signal + repair operator, and the finding that a plain supervised net matches on accuracy but not consistency — so the value is reliability, not accuracy (§5.3). (3) Test-time energy repair framed as inference-time compute, with a measured compute–consistency curve (§5.2). (4) The truth-propagation wave: reasoning depth $\mapsto$ test-time compute is linear, and the t-norm sets the speed (§5.1). (5) A small product — `score() / repair()` — and a worked config-validation use case.

2. Method

Throughout, an **atom** is a single atomic proposition — one true/false statement such as `Rain` or `sso_enabled` (the standard logic term for the smallest unit of a rule; unrelated to physics atoms despite the thermodynamic framing).

Let $p_i \in [0,1]$ be the belief that atom i is true. A literal's truth is p_i (or $1-p_i$ if negated). A Horn rule $L_1 \wedge \dots \wedge L_k \rightarrow C$ scores as $\text{sat} = \text{residuum}(\otimes_i \text{truth}(L_i), \text{truth}(C))$. Modus ponens is emergent: with the antecedent believed, $\text{residuum}(1, p_C) = p_C$, so $\text{sat}=1$ requires $p_C=1$. Per-rule energy is $E_r = \exp(\beta(1-\text{sat}_r)) - 1$; total $E = \text{mean}_r E_r$. Valid $\Rightarrow E=0$; contradictions grow exponentially with inverse temperature β .

Table 1 — Operator mapping per t-norm family.

Discrete	Product	Łukasiewicz (default)	Gödel
$\neg a$	$1-a$	$1-a$	$1-a$
$a \wedge b$	$a \cdot b$	$\max(0, a+b-1)$	$\min(a,b)$
$a \Rightarrow c$ (residuum)	$1 \text{ if } a \leq c \text{ else } c/a$	$\min(1, 1-a+c)$	$1 \text{ if } a \leq c \text{ else } c$

Test-time repair (the key operator). Given any belief state p (e.g. a model's output on a novel input), hold trusted atoms fixed and run gradient descent on E over the rest, in logit space: $p^* = \text{argmin } E(p)$. Because minimizing E enforces every rule, this repairs an inconsistent output to the nearest valid state; the number of steps is a compute budget.

3. Benchmark

The expanded knowledge base has 11 atoms and 9 rules, yielding 18 distinct logically-consistent worlds (294 of 2048 full Boolean worlds are consistent). Five atoms are observed causes (`Rain`, `Sprinkler`, `Cold`, `HeaterOn`, `Windy`); six are derived (`Cloudy`, `WetGround`, `Slippery`, `Ice`, `Warm`, `Chilly`) via rules including two-literal antecedents (`Cold $\wedge$ WetGround $\rightarrow$ Ice`) and a two-hop chain (`Rain $\rightarrow$ WetGround $\rightarrow$ Slippery`), plus negative constraints (`Rain $\rightarrow$  $\neg$ Sprinkler`). We hold out entire worlds: train on 12, test on 6 never seen. Inputs are noisy encodings of the causes ($\sigma=0.12$) with 3 distractor dimensions.

4. Experiments

4.1 Logical energy as test-time compute (the propagation wave)

Starting from a state where a cause is true but a chain of derived atoms is false, test-time energy descent (plain SGD) turns atoms on one hop at a time — atom depth d activates only after $d-1$. Activation step is a strictly monotone, near-linear function of depth; the t -norm sets the slope (Łukasiewicz 3.4 steps/hop; Product/Gödel ~ 1.0). The clean monotone wave requires momentum-free SGD; the robust claim is the within-chain wave and linear depth-cost at fixed length. We stress that this linear scaling is *intuitive, not surprising* — any iterative fixed-point or message-passing procedure needs steps proportional to how far information must travel. The contribution is a clean, visual *demonstration* of that behaviour in a differentiable-logic energy, not the discovery of a new scaling law.

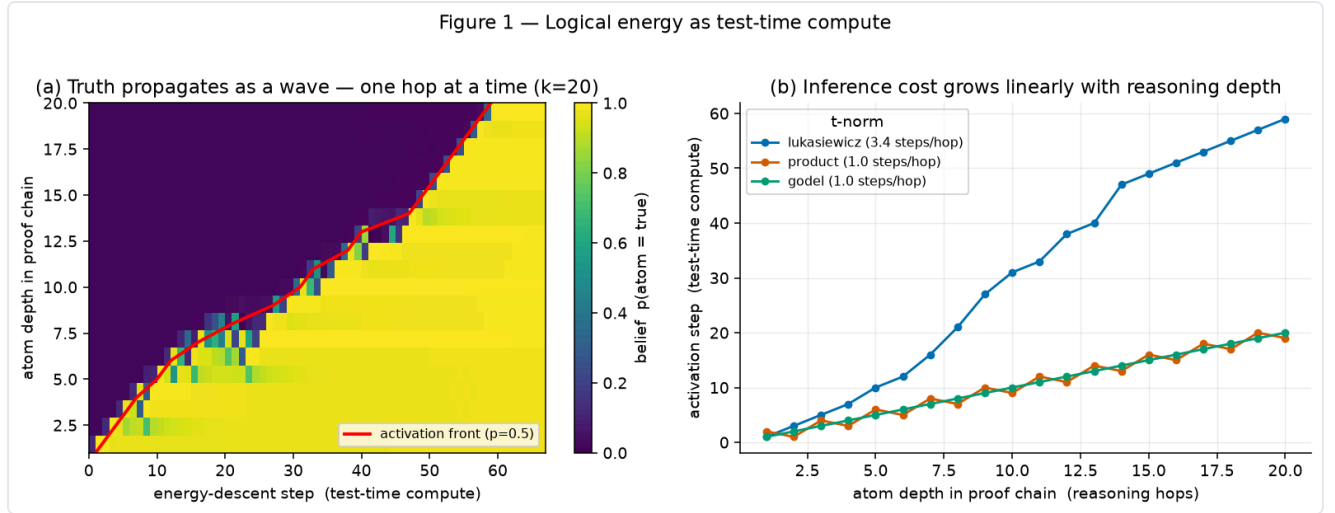


Figure 1. (a) Truth propagates as a wave — one hop at a time. (b) Inference cost grows linearly with reasoning depth; the t -norm sets the propagation speed.

4.2 Generalization and test-time repair

On worlds never seen in training, a single feed-forward pass generalizes only partially and leaves outputs logically inconsistent (high energy). Repair fixes them, and is a compute knob: more steps $\rightarrow$ lower energy and higher accuracy, plateauing near 50 steps.

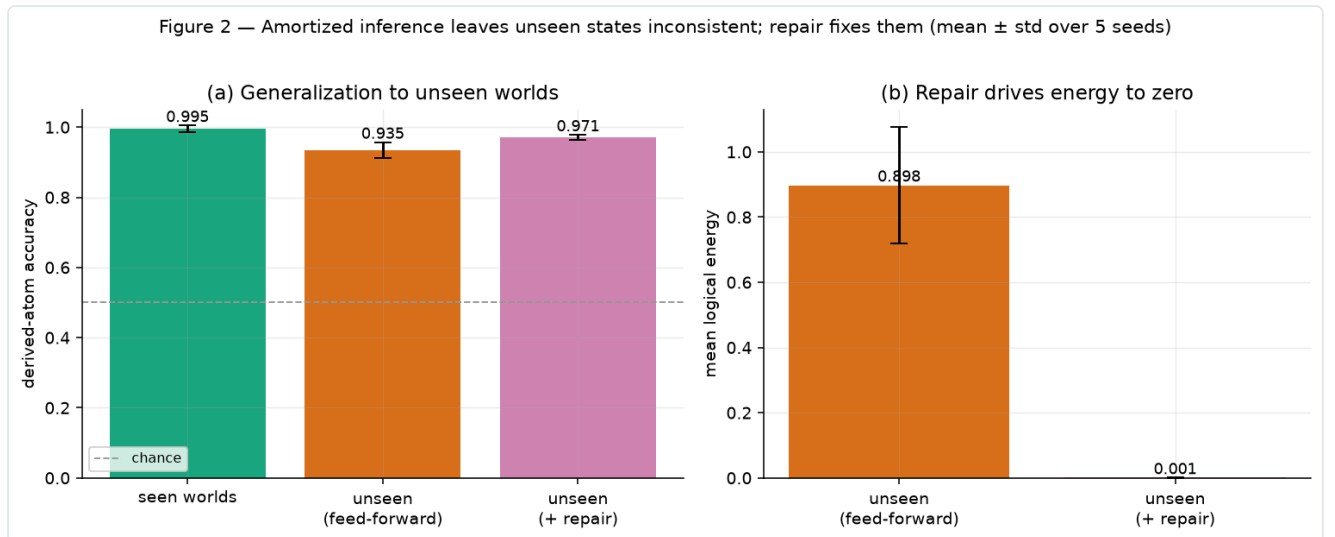


Figure 2. Amortized inference leaves unseen states inconsistent; repair drives energy to zero (mean $\pm$ std, 5 seeds).

Table 2 — Generalization (mean $\pm$ std, 5 seeds).

Split	Derived accuracy	Mean energy
Seen worlds	0.995 ± 0.010	~ 0
Unseen (feed-forward)	0.935 ± 0.022	0.898 ± 0.178
Unseen (+ repair)	0.971 ± 0.007	0.001 ± 0.000
Cause recovery (unseen)	0.981 ± 0.004	—

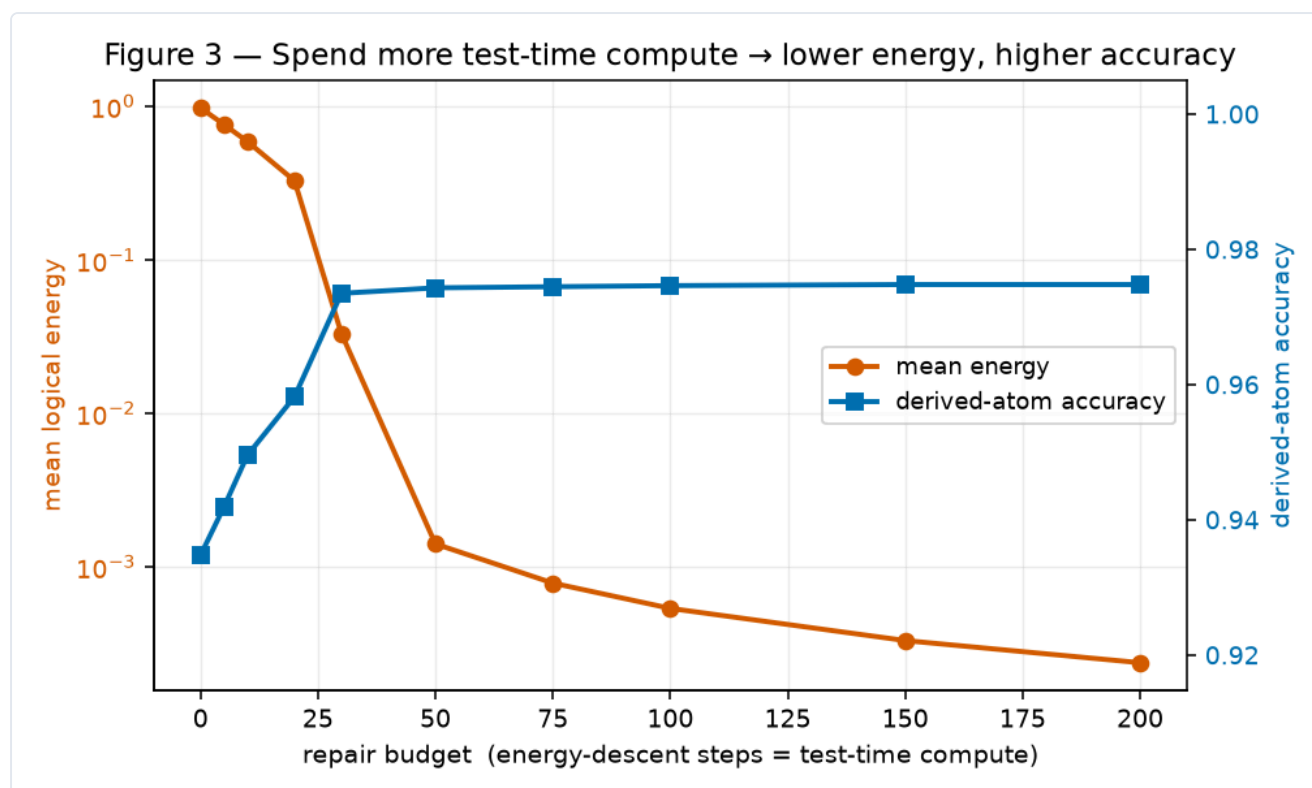


Figure 3. Test-time compute vs. energy and accuracy: spend more descent steps $\rightarrow$ lower energy, higher accuracy.

4.3 Baselines — the value is consistency, not accuracy

A plain supervised network (BCE on all atoms, no logic) reaches 0.955 accuracy on unseen worlds — comparable to ThermoLogic + repair (0.970). But its outputs are logically inconsistent (energy 0.946), with no signal that anything is wrong. Only the energy provides a label-free consistency guarantee (0.001). **This is the honest headline: ThermoLogic is a reliability mechanism, not a way to beat a classifier on accuracy.**

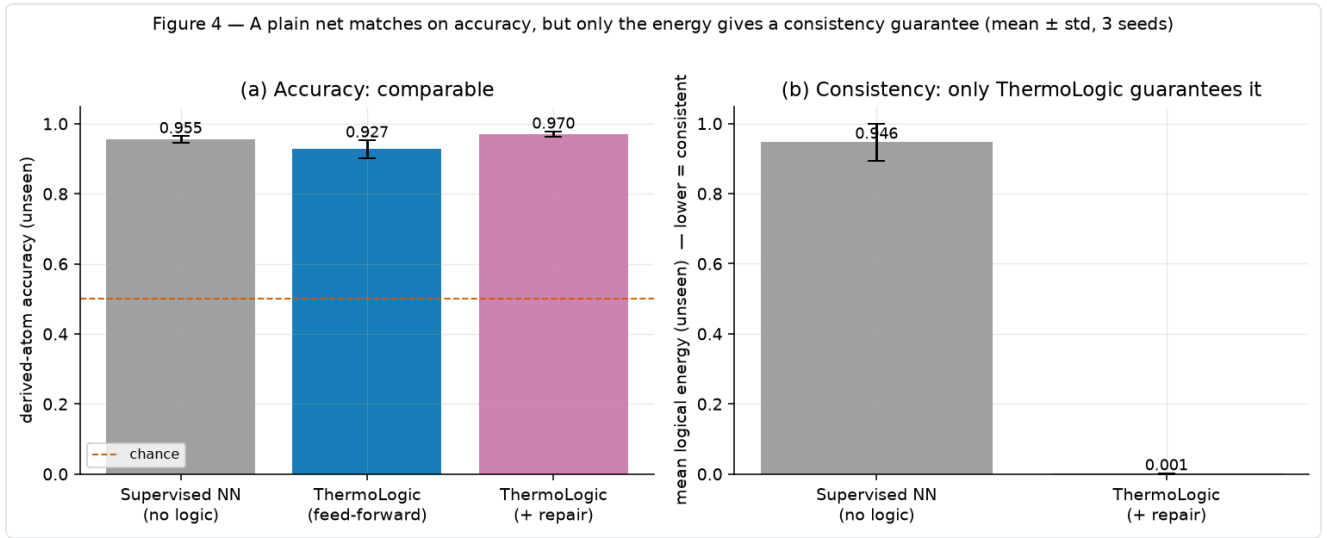


Figure 4. Comparable accuracy; only the energy guarantees consistency (mean $\pm$ std, 3 seeds).

Table 3 — Baselines on unseen worlds (mean $\pm$ std, 3 seeds).

Method	Derived accuracy	Mean energy (consistency)
Supervised NN (no logic)	0.955 $\pm$ 0.010	0.946 $\pm$ 0.053 $\times$ inconsistent
ThermoLogic (feed-forward)	0.927 $\pm$ 0.026	0.90
ThermoLogic (+ repair)	0.970 $\pm$ 0.008	0.001 $\pm$ 0.000 $\checkmark$

4.4 Robustness and t-norm comparison

Repair helps most when causes are recoverable (low–moderate noise); at high noise ($\sigma \geq 0.3$) mis-recovered causes cap accuracy. As more worlds are held out, generalization degrades gracefully. Among t-norms, Łukasiewicz's smoother gradients give the best accuracy and repairability (0.970 vs 0.876 Product, 0.878 Gödel) — the default.

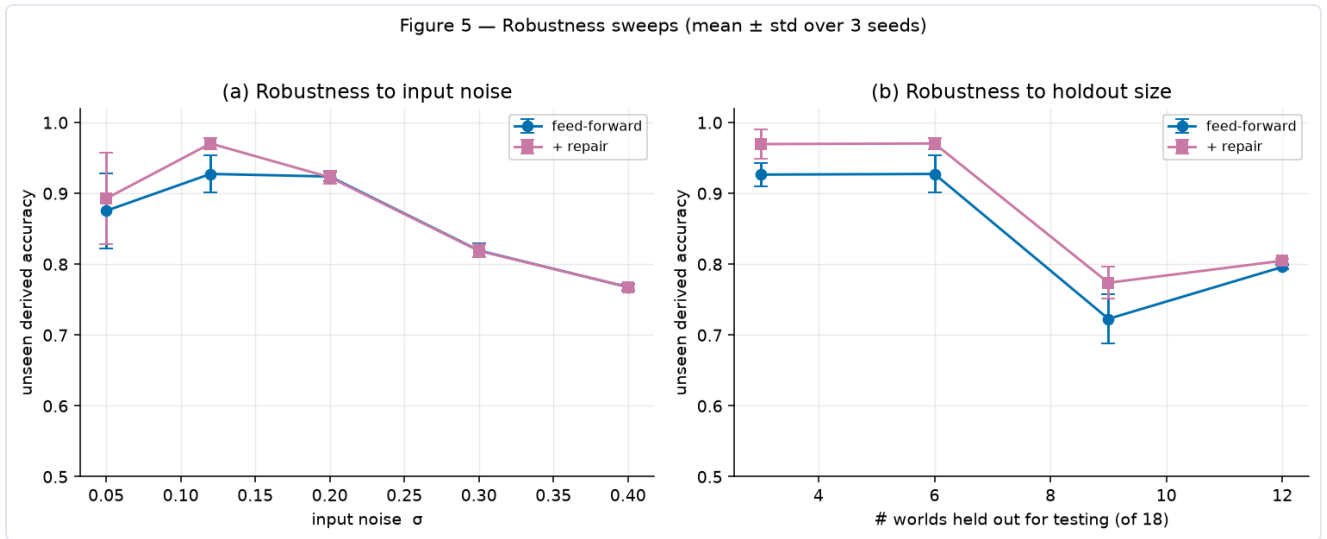


Figure 5. Robustness to input noise (a) and holdout size (b); mean $\pm$ std, 3 seeds.

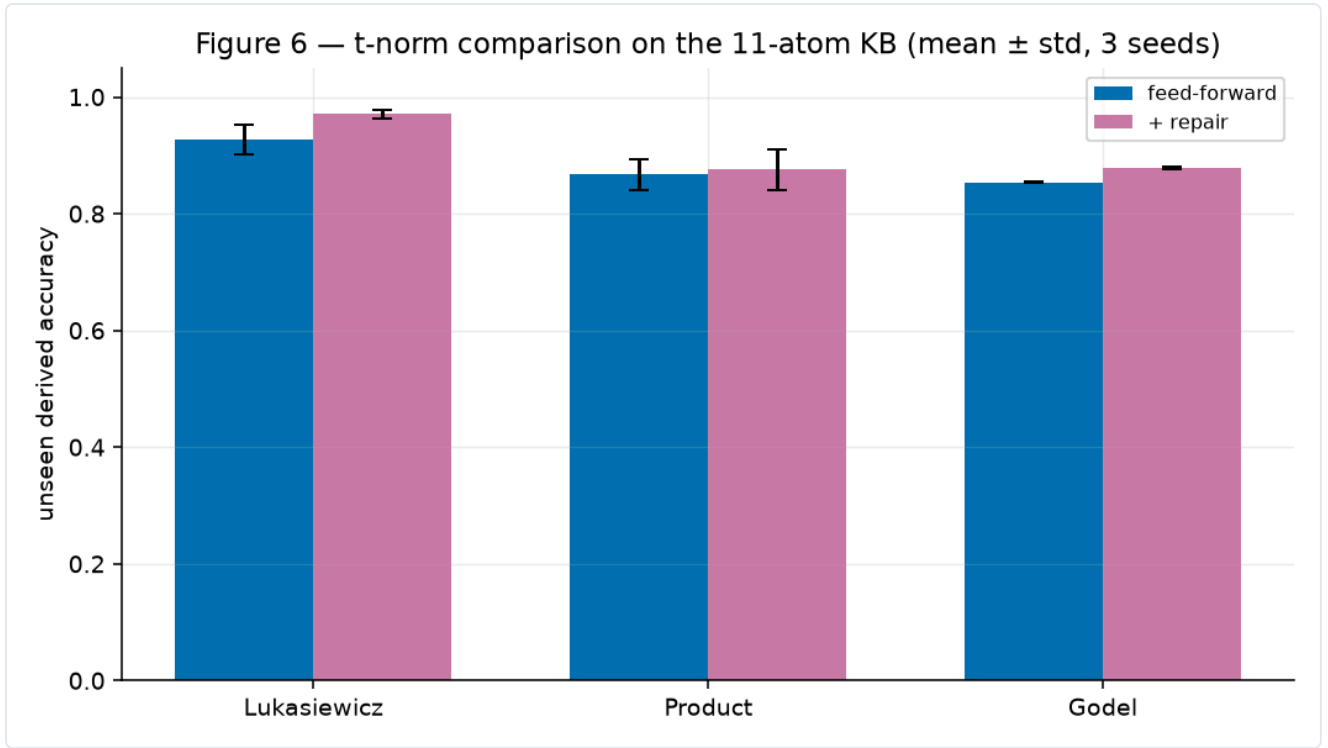


Figure 6. t-norm comparison on the 11-atom KB.

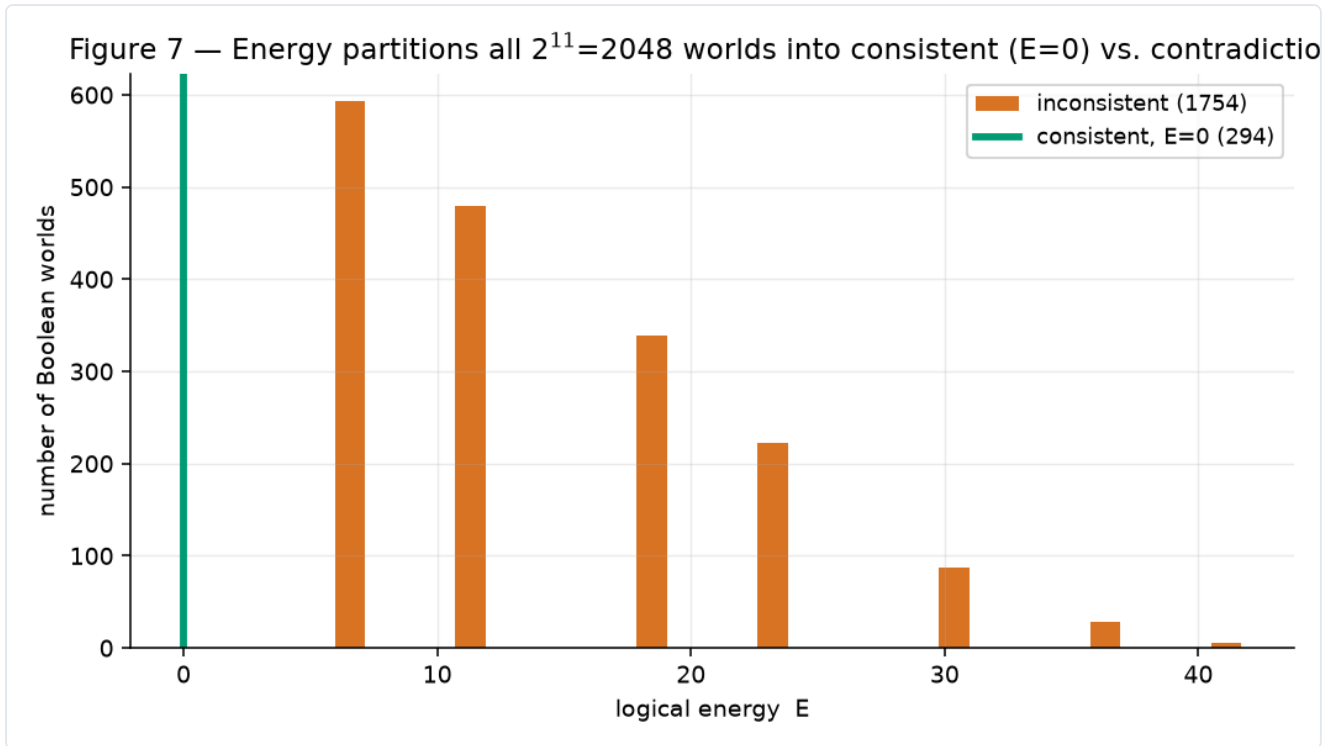


Figure 7. Energy partitions all $2^{11} = 2048$ Boolean worlds into consistent ($E=0$) vs. contradictions (294 consistent; the rest carry energy in $[5.96, 41.69]$).

4.5 Do you even need learning? Energy repair vs. projection

The obvious objection is “why not just project the output onto the nearest valid state with a solver?” We compare against forward-chaining (a Horn-KB oracle), an exact confidence-weighted projection (enumerate valid completions, pick the nearest to the model’s soft output), and a greedy flip search — all holding the recovered causes fixed. **We report this honestly: on accuracy, energy repair does not win.** Exact projection and the forward-chaining oracle match it (0.969–0.973 vs 0.970, within noise), and the exact projection changes the fewest bits by

construction; energy repair lands on the *same* valid state as the exact projection 96% of the time. Its advantages are elsewhere and real: **(i) scaling** — projection enumerates 2^d completions (0.4 ms at $d=4 \rightarrow 968$ ms at $d=18$, crossing energy repair's flat cost near $d \approx 17$; MaxSAT/ILP is also NP-hard and non-differentiable), while energy repair stays $O(\text{steps} \times \text{rules})$; and **(ii) differentiability** — a projection is a non-differentiable argmin, whereas repair can be trained through.

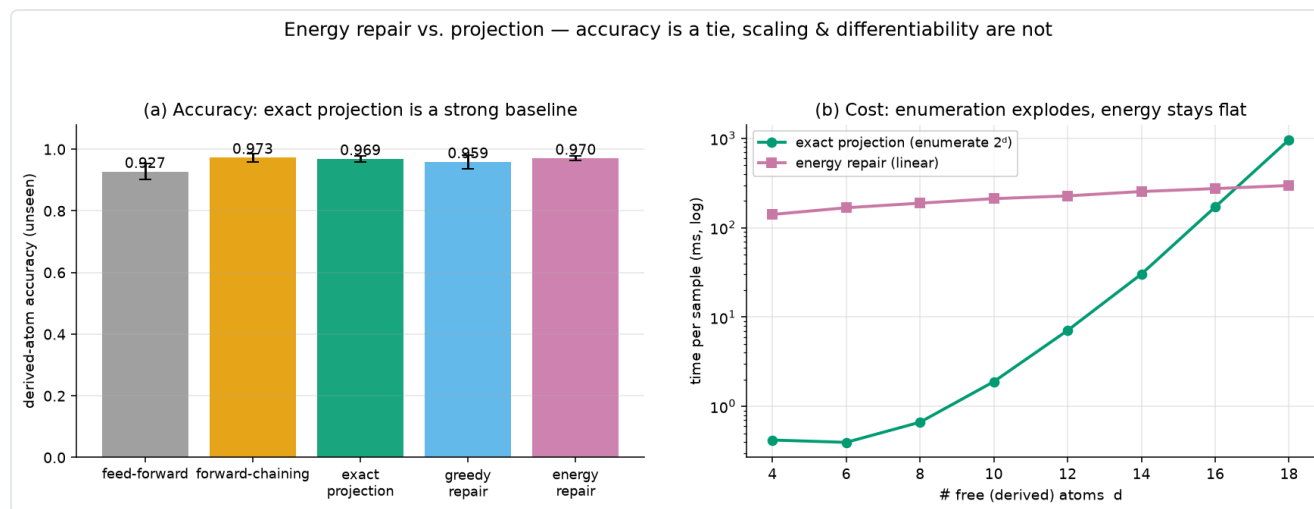


Figure 8. Energy repair vs. projection: accuracy is a tie (a); scaling and differentiability are not (b).

Training through repair (the capability projection lacks). Because repair is gradient descent, it can be unrolled into a differentiable module and placed inside a training loop. The gradient of a post-repair loss w.r.t. the network is **0.19 through differentiable repair and exactly 0.00 through a projection**. Supervising a proposer with *only a scalar readout* of the repaired state — no per-atom labels — and training end-to-end through the repair, per-atom logical correctness emerges: derived-atom accuracy climbs from 0.47 to **0.993**. A projection-in-the-loop gets zero gradient and cannot learn this. That is the honest answer to “why not just project?”: when the repair must live inside a learned system, differentiability is the whole point.

A negative result (for balance). We also tested whether training through the repair is a *general* inductive-bias win. With full per-atom labels and identical settings, pre-repair and post-repair supervision tie exactly on held-out worlds (e.g. 0.807 vs 0.805 at 6 training worlds; 0.948 vs 0.948 at 12) — no benefit, because the derived logic here is easily learned from data. Logic-in-the-loop helps only when the relationship is hard to learn; we do not claim a generalization win, only the distinct *capability* above.

...but it helps where theory predicts. Used as a semi-supervised loss on *unlabelled* data for a *hard* target (parity of four bits, which MLPs generalize poorly), the differentiable-logic energy lifts feed-forward parity accuracy by up to **+26 points** over labels-only (0.65 $\rightarrow$ 0.91 at 64 labels), with the gain largest when labels are scarce and vanishing once labels are plentiful. The precise boundary: the logic is inert when the target is easy and labels plentiful, and valuable when the target is hard and labels are few.

Is it just Semantic Loss? A logic loss on unlabelled data is what Xu et al.'s **Semantic Loss** (ICML 2018) already does, as $-\log$ of the exact weighted model count. We ran three arms on the identical parity task (labels-only, + our fuzzy energy, + exact Semantic Loss). They are *not* the same object: at matched weight the fuzzy energy is stronger (0.909 vs 0.831 vs 0.651 labels-only at 64 labels), and sweeping the loss weight shows why the gap is not an artifact — the fuzzy energy is weight-robust (0.90–0.92 for $w \in [0.5, 4]$) while Semantic Loss peaks at a small weight and collapses to 0.57 as the weight grows. We claim no general superiority (one task), but the “merely a re-derivation” concern is retired and needing no weight tuning is a real convenience.

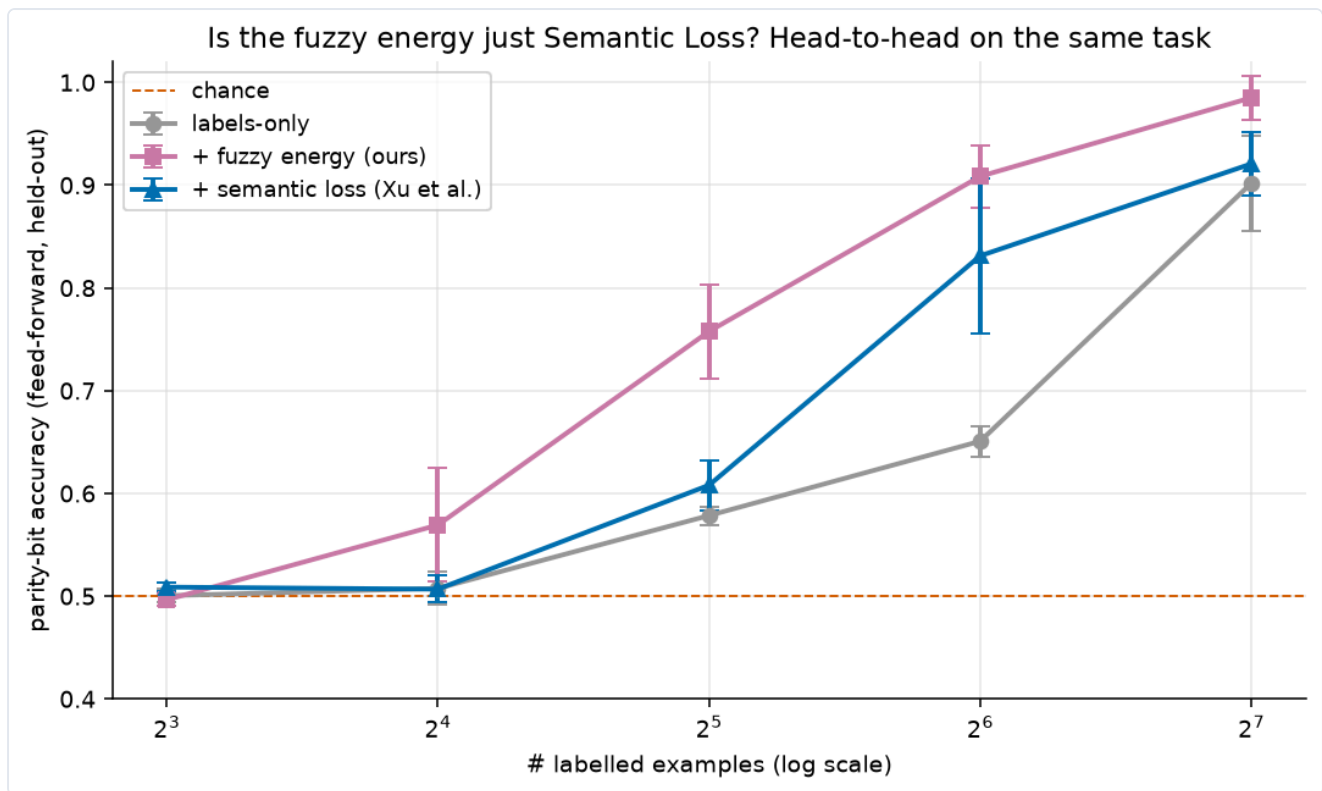


Figure 9. Novelty check: the fuzzy energy is a distinct mechanism from Semantic Loss and, on this semi-supervised parity task at matched weight, at least as accurate — and far more robust to the loss weight.

5. The product

The mechanism ships as `thermologic`, a differentiable logic-energy layer that drops after any model: `score()` gives a label-free inconsistency signal, `violations()` names the broken rules, and `repair(budget=...)` returns the nearest valid output at a controllable compute budget. A worked SaaS config-validation example flags an invalid plan (e.g. a Pro request for enterprise-only SSO), names the rule, and repairs to the nearest valid configuration. v1 scope: structured/tabular outputs with propositional rules; general LLM-text validation is roadmap, not a claim.

Harder benchmark + end-to-end integration. A realistic cloud security-configuration benchmark, written as first-order templates ($\forall$ resource r) and *grounded* over 3 resources, gives 20 atoms, 19 rules, and **432 distinct worlds** — the pipeline reaches 100% consistency there. Wiring `LogicEnergy` onto a *plain* network trained with no logic, it flags the 13.6% of unseen configs that violate policy (exact rule, no labels; ~100% precision) and repairs to 100% consistency. Honest limits: it detects *policy inconsistency*, not general correctness, and repair guarantees consistency (not correctness) against the model's recovered inputs.

An adversarial head-to-head — where our method loses. To rebut the “circular ground truth” objection we ran a deliberately hostile test on a canonical, externally-defined problem: recovering order-4 **Latin squares** (labels = the 576 real squares, enumerated independently; 64 atoms, 288 rules) from a noisy observation. It loses on *both* axes, and we report it plainly. At runtime an exact nearest-valid-grid solver **dominates** soft energy repair (100% vs 3.6% valid; 0.27 vs 0.03 grid-exact) — for small, clean, propositional constraints a solver is simply the right tool. At training time the logic loss gives *no* lift (a shade lower at every label budget), because Latin-square constraints are permutation-symmetric: they forbid conflicts but carry no signal about *which* value belongs in a cell. This sharpens the semi-supervised boundary above: the logic aids learning only when the constraint is *informative about the target* (parity), not merely satisfied by it.

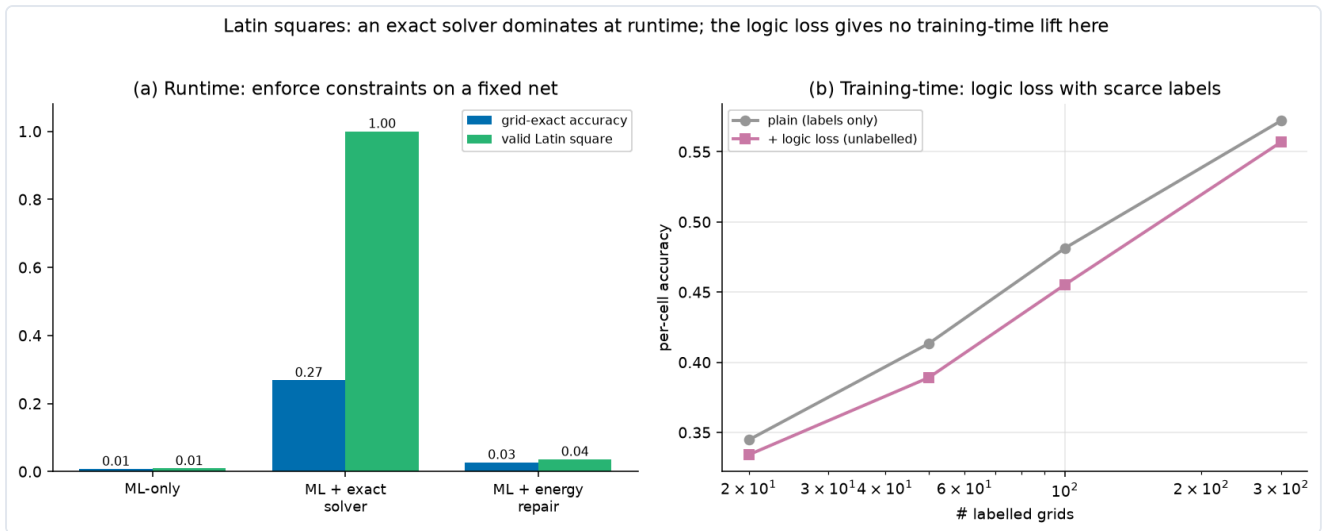


Figure 10. The honest external head-to-head on Latin squares. (a) At runtime an exact solver dominates soft energy repair; (b) at training time the logic loss gives no lift on this permutation-symmetric task. The method wins when you need a gradient, scale past enumeration, or scarce-label target signal — not here.

When enumeration is infeasible — measured scaling. The flip side of the Latin loss: when the number of valid states is large, the enumerate-and-project baseline dies while energy repair keeps running. Sweeping Latin squares of order 3/4/5 ($12 \rightarrow 576 \rightarrow 161\,280$ valid grids) up to **9×9 Sudoku** ($\sim 6.67 \times 10^{21}$ valid grids), the projection cost grows ~ 20 orders of magnitude while energy-repair wall-clock grows $\sim 30\times$ ($1.2s \rightarrow 34.8s$). Honest caveat kept prominent: this is *not* “solvers fail” — a real CP/SAT solver still cracks Sudoku instantly; what dies is the enumerate-and-project baseline, and the differentiable energy is the generic, gradient-compatible fallback that survives. A model-agnostic guardrail on LLM-style JSON (`examples/llm_json_guardrail.py`) makes the product concrete: it repairs only *safety* settings, holds user *intent* fixed, and escalates genuine intent conflicts (a public bucket that also stores PII) to a human rather than silently rewriting the request.

Soft penalty → hard guarantee. The sharpest self-criticism was that soft energy makes contradictions *expensive*, *not impossible* — repair could return a still-invalid state. A discrete backstop (min-conflicts local search) closes it: `repair(guarantee=True)` returns a **verified** rule-valid state or raises, and when the request is self-contradictory it returns a **proof** (the conflicting rules) instead of failing silently. Run on five configs generated by Claude in-session, the guard flags four, verifiably repairs three, and proves one — a public dashboard over raw customer PII — unsatisfiable.

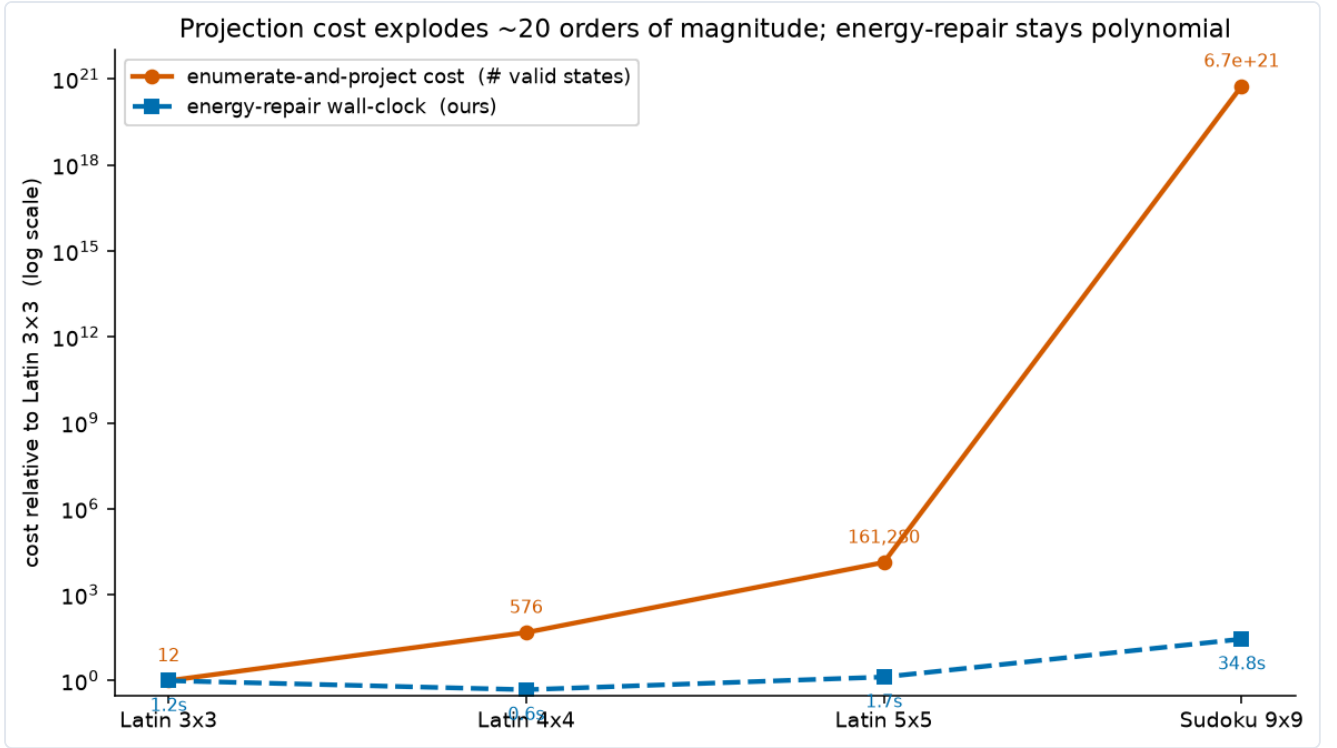


Figure 11. Measured scaling: the enumerate-and-project cost explodes ~20 orders of magnitude from a 3×3 Latin square to 9×9 Sudoku, while energy-repair wall-clock stays polynomial (~30×).

6. Discussion & limitations

The honest value proposition is a label-free consistency guarantee and repair, not beating supervised learning on accuracy. The energy adds an iterative, budgetable inference step — a small instance of test-time compute for logical consistency. Limitations: 11 atoms, propositional Horn-style rules, synthetic data; no first-order variables, no defeasible rules, no real LLM. Soft energy makes contradictions expensive, not impossible (a crisp projection step would give a hard guarantee). The clean propagation wave needs momentum-free SGD, and mean-energy normalization couples wave speed to chain length.

7. Conclusion

A differentiable fuzzy-logic energy lets a neural network's beliefs be scored and repaired against hard rules by backpropagation alone. Empirically the value is reliability: comparable accuracy to a plain net but with a label-free consistency guarantee, an inference-time repair that scales with a compute budget, and a clean picture of reasoning depth as test-time compute — a small, honest, reproducible substrate for constraint-aware ML.

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References are indicative of the intellectual context; this is a self-contained prototype, not a comparative benchmark against these systems. Code, figures, and this report regenerate from the open repository with fixed seeds.